

OneGrow: Unified Temporal Plant Image and Mask Generation

Supplementary Material

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Section S1 gives the training objective, the architecture and conditioning, and the data and optimization settings. Section S2 reports mask-to-image generation quality. Section S3 shows the temporal roll-out and the reveal-defined tasks on both datasets. Terminology, symbols, and metrics follow the extended abstract.

S1 Implementation Details

Training objective. A window is indexed by views $\mathcal{V} \subseteq \{0, \dots, M-1\} \times \{0, \dots, F-1\}$, where the first index is the modality, with $m=0$ for images and $m=1$ for masks, and the second index is the frame. A full paired window has $M=2$ modalities and $V=|\mathcal{V}| = MF$ views. A view v holds a latent $z_v \in \mathbb{R}^{C \times h \times w}$, split into P non-overlapping $p \times p$ patches, where C denotes the number of latent channels and h and w denote the downsampled latent height and width. The model regresses the velocity target of generated patches under the reveal-weighted mean squared error (MSE)

$$\mathcal{L}(\theta) = \mathbb{E} \left[\frac{\sum_{v \in \mathcal{V}, j} w_{v,j} \|u_\theta(z_\tau, \tau, \mathbf{q})_{v,j} - (z_{v,j} - \epsilon_{v,j})\|_2^2}{\max\left(1, \sum_{v \in \mathcal{V}, j} w_{v,j}\right)} \right], \quad w_{v,j} = 1 - r_{v,j}, \quad (\text{S1})$$

where $\mathbf{q}$ denotes the conditioning tokens, r denotes the reveal, and the expectation is over the data, the noise ϵ , the reveal, and the shared flow time $\tilde{\tau}$. The weight $w_{v,j}$ is zero on revealed patches, so only generated patches contribute.

Architecture. Each latent patch becomes a token augmented with its modality and frame coordinates. The transformer applies rotary position embedding (RoPE) over modality, time, and spatial coordinates [15], while the flow time enters through adaptive layer-normalization modulation [10]. Each block applies query-key normalization for training stability [4] and uses SwiGLU feed-forward layers [13]. A shared linear head maps tokens back to velocity patches. The transformer has width 512, 16 blocks, 8 attention heads, and latent patch size 2, giving $P=1024$ patches per view. Both modalities use the frozen SD3.5 variational autoencoder (VAE) [5] with 16 latent channels and eightfold downsampling, so a 512×512 image becomes a 64×64 latent.

Conditioning. Beyond the reveal context, the model accepts optional per-window metadata. Each conditioner is represented by a *register token* with a learned null embedding. The reported model conditions on the acquisition *domain* and on per-frame *photometric* statistics for exposure and color. Conditioners are dropped independently during training for classifier-free guidance (CFG) [6]. Frames without a known acquisition time use a learned null-time embedding.

Reveal sampling and view dropout. During training, reveal configurations are drawn from a fixed non-uniform distribution over task families that emphasizes the target tasks of mask-conditioned synthesis and forecasting while still covering the full range of image-mask relationships. Training also applies per-batch *view dropout*, which drops a shared random subset of views, so the model sees variable and sometimes unpaired image and mask context. Validation and evaluation use the complete available view set.

Data. GWFSS [16] released 99 images with human organ masks for training and about 64k unlabeled images for pretraining, and the 1,096-image challenge test set was made public with human masks after the challenge closed. The pseudo-labels come from the winning challenge entry, a ViT-Adapter and Mask2Former network with a masked-image-modeling backbone [1–3,11]. For FIP1, that model runs tile-wise and blends per-class scores. FIP1 frames are natively 960×640 . OneGrow takes a plot-aligned 256×256 inner-plot crop of each frame and up-samples it to 512×512 before encoding, with windows of $F=4$ frames. The two sources are sampled with equal probability. FIP1 shares an acquisition domain with one GWFSS institution, which links the two sources through the domain conditioner.

Optimization and sampling. We optimize Eq. (S1) with AdamW [8], using $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, a peak learning rate of 3×10^{-4} , linear warm-up, cosine decay, gradient clipping, and an exponential moving average for evaluation. Unless noted, sampling uses a midpoint ordinary differential equation (ODE) solver with 50 steps and guidance scale 2.

S2 Mask-to-Image Generation Quality

Table S1 reports the mask-to-image numbers summarized in the extended abstract, together with the FIP1 scores the abstract omits. Metrics are the Fréchet distance in DINOv2 feature space (FD-DINOv2) [9,14], the CLIP maximum mean discrepancy (CMMD) [7,12], and paired Learned Perceptual Image Patch Similarity (LPIPS) [17], chosen for small reference sets and for imagery outside the ImageNet domain.

Table S1: Mask-to-image generation quality on GWFSS and FIP1, against held-out reference imagery. Lower is better. *Mask-NN* is a training-free retrieval baseline that returns, for each test mask, the real training image whose mask latent is nearest. It conditions on the mask only, retrieves from the training split, and is scored against the held-out set. Mask-NN returns real images, which favors the two set-level metrics, while paired LPIPS rewards matching the particular held-out frame. GWFSS scores all 1,096 held-out images and FIP1 scores 250 held-out windows, so the set-level metrics carry more sampling noise on FIP1.

Dataset	Method	FD-DINOv2 ↓	CMMD ↓	LPIPS ↓
GWFSS	Mask-NN	257.009	15.968	0.548
	OneGrow (mask-to-image)	322.168	19.332	0.426
FIP1	Mask-NN	187.404	26.039	0.478
	OneGrow (mask-to-image)	189.640	9.564	0.333

S3 Qualitative Results

Throughout this section, content outlined in blue is revealed to the model as clean context and the remaining content is generated. Most figures place the reference beside the generation inside a single *cell*. In those figures every generated cell is split down the middle, with the held-out reference in the left half and the generated content in the right half. A row described below as generated is therefore half reference and half generation, not generation alone. The generated half is a sample and need not match the reference exactly. Figures S1 and S9 use no split cells, and their captions state what each row and panel shows.

S3.1 Temporal roll-out

Figure S1 shows the sliding-window roll-out that generates an image sequence from a mask sequence beyond the four-frame training window.

S3.2 Reveal-defined tasks on FIP1 growth windows

Figures S2 to S6 show five reveal configurations of the same trained model on FIP1 windows. Columns f_0 to f_3 are frames of one plot, ordered by acquisition time. These early-season windows contain two classes, background (black) and leaf (yellow-green).

S3.3 Reveal-defined tasks on GWFSS images

GWFSS contributes single-frame windows, so the same reveal configurations apply to one frame at a time (Figs. S7 and S8). These images are later-stage canopies and contain stem pixels (blue) alongside background (black) and leaf (yellow-green).

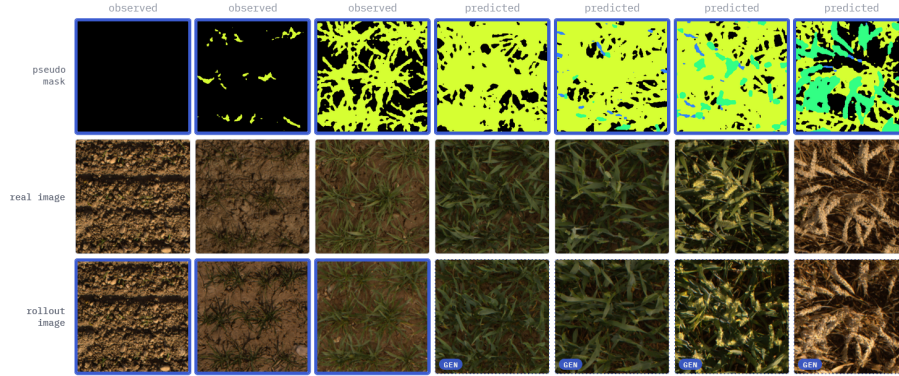


Fig. S1: Qualitative temporal roll-out on a FIP1 growth sequence, one column per time step, ordered left to right. *Top:* the teacher pseudo-mask, revealed for every frame (blue border). *Middle:* the real image, shown for reference. *Bottom:* the roll-out. The first three frames (*observed*, blue border) are real images that seed the roll-out. Each later frame (*predicted*, marked GEN) is generated from its mask and the three preceding frames, so generated images re-enter as context as the window slides. Mask colors: background (black), leaf (yellow-green), stem (blue), head (green).

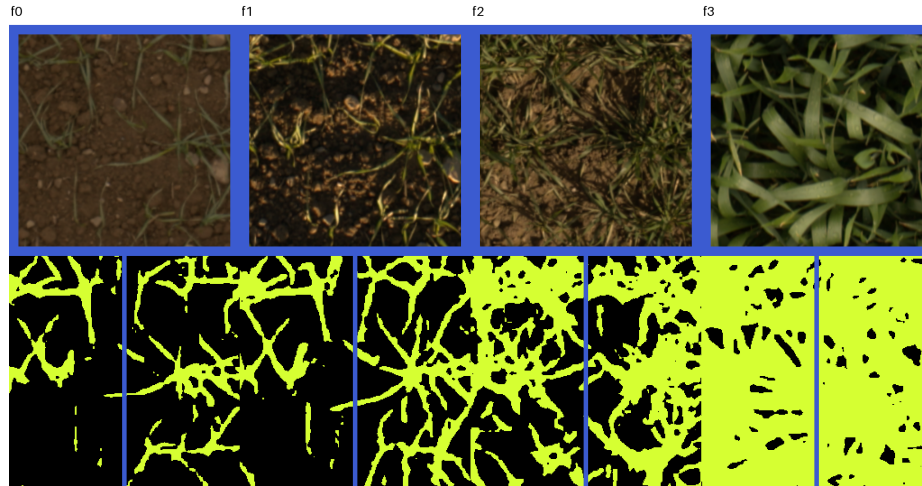


Fig. S2: Image-to-mask segmentation on a FIP1 growth window. The image views (*top row*) are revealed as clean context, and OneGrow generates the organ masks (*bottom row*). Each bottom cell is split down the middle into the held-out reference mask (left half) and the generated mask (right half).

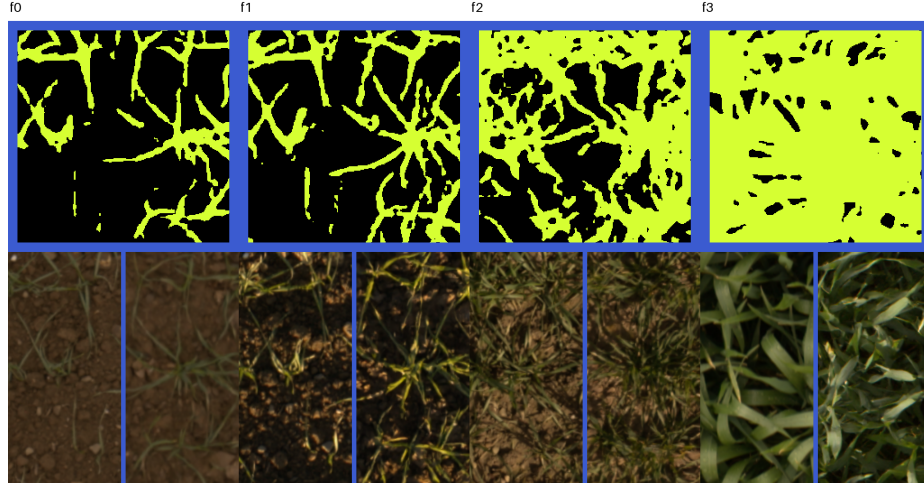


Fig. S3: Mask-to-image synthesis on a FIP1 growth window. The organ masks (*top row*) are revealed as clean context, and OneGrow generates the images (*bottom row*). Each bottom cell is split down the middle into the held-out reference image (left half) and the generated image (right half). The seam falls mid-cell, so continuity between the two halves is visible.

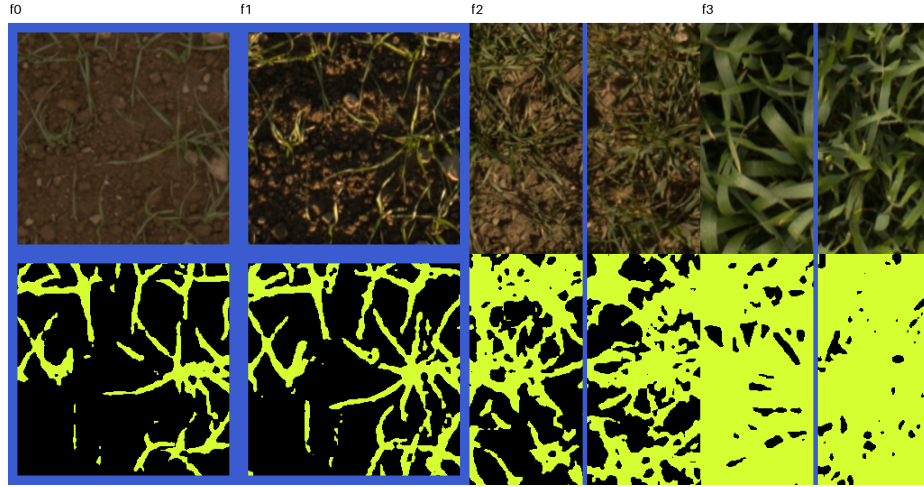


Fig. S4: Temporal forecasting on a FIP1 growth window. The image (*top row*) and mask (*bottom row*) of the first two frames are revealed as clean context, and OneGrow forecasts the later frames. Each forecast cell is split down the middle into the held-out reference (left half) and the forecast (right half).

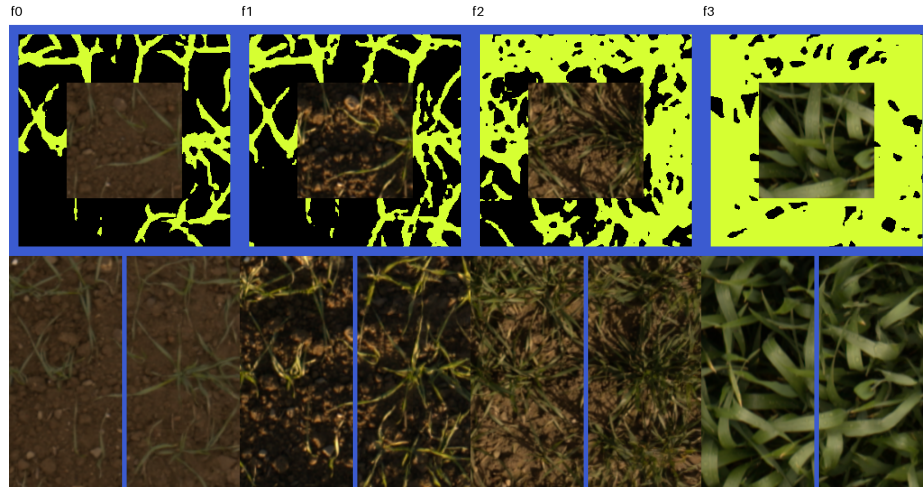


Fig. S5: Mask-guided spatial completion on a FIP1 growth window. The full organ mask with a central image patch composited into its center (*top row*) is revealed as clean context, and OneGrow outpaints the surrounding image (*bottom row*). Each bottom cell is split down the middle into the held-out reference image (left half) and the outpainted image (right half).

S3.4 Progressive spatial outpainting

The spatial-completion reveal also lets the model grow an image beyond its native window size (Fig. S9). Starting from one real image-mask tile, half-overlapping windows are outpainted outward and assembled into a larger canvas of both modalities. Each new window reveals its overlap with the already-generated canvas, so every seam falls on revealed content.

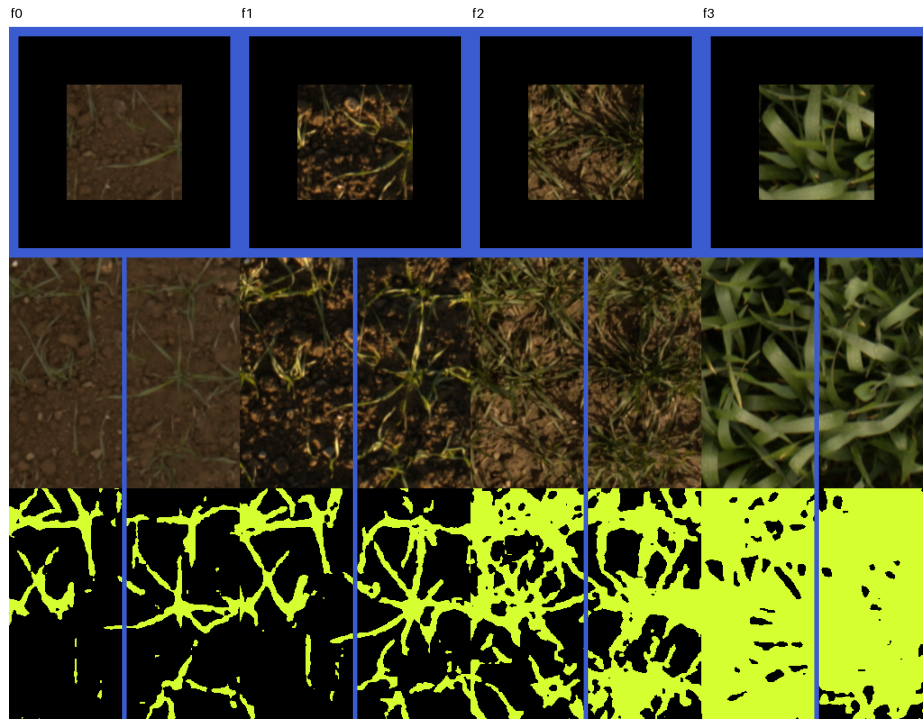


Fig. S6: Joint spatial completion on a FIP1 growth window. Only a central image patch is revealed (*top row*), and OneGrow jointly generates the surrounding image (*middle row*) and the full organ mask (*bottom row*). Cells in the two generated rows are split down the middle into the held-out reference (left half) and the generated content (right half). No mask is given here, which distinguishes this task from Fig. S5.

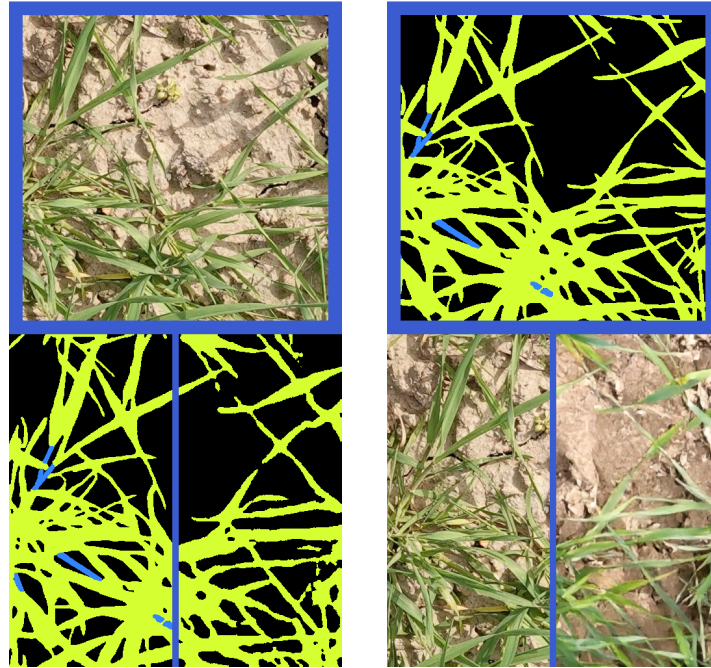


Fig. S7: Single-frame tasks on GWFSS. *Left panel:* image-to-mask segmentation, where the image is revealed (top row) and the organ mask is generated (bottom row). *Right panel:* mask-to-image synthesis, where the mask is revealed (top row) and the image is generated (bottom row). Each generated cell is split down the middle into the held-out reference (left half) and the generated content (right half).

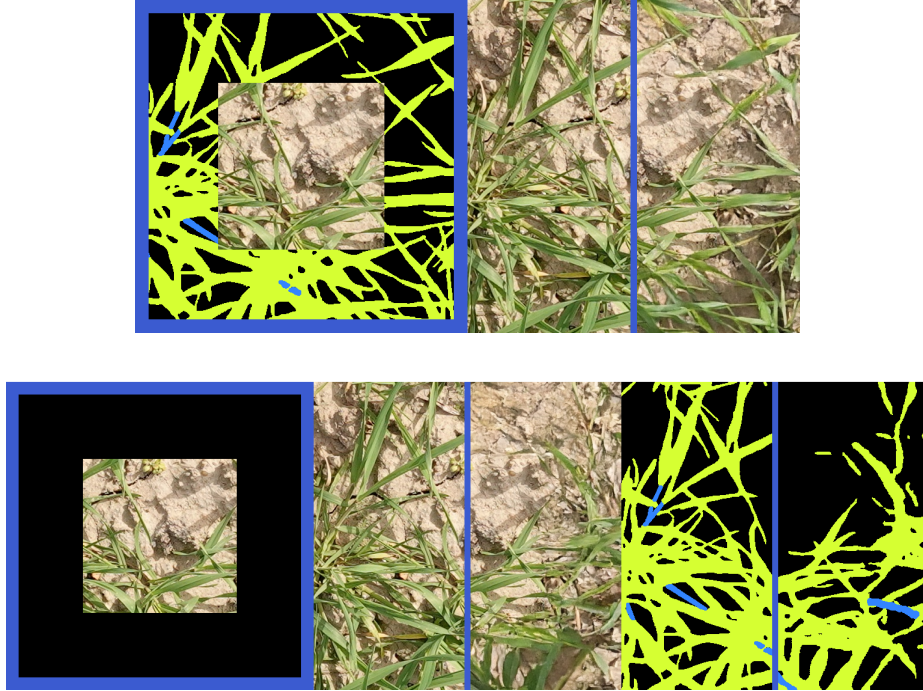


Fig. S8: Spatial completion on GWFSS. Cells run left to right within each row. *Top:* mask-guided completion, where the full organ mask with a central image patch composited into its center is revealed (first cell) and the surrounding image is generated (second cell). *Bottom:* joint completion, where only the central image patch is revealed (first cell) and OneGrow generates both the surrounding image (second cell) and the organ mask (third cell). Each generated cell is split down the middle into the held-out reference (left half) and the generated content (right half).

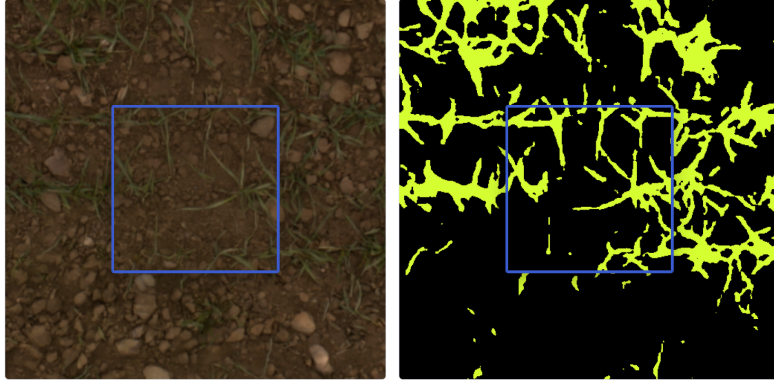


Fig. S9: Progressive spatial outpainting from a single seed tile. Starting from one real image-mask tile (outlined in blue), OneGrow grows a larger canvas by iterated half-overlap outpainting, jointly generating the surrounding image (*left panel*) and organ mask (*right panel*). Each panel is one assembled canvas and is not split into reference and generated halves. Each outward window is conditioned on its overlap with the already-generated canvas, and the fully-convolutional decoder renders the assembled canvas in a single pass.

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